Questions by topic: Captive breeding, habitat creation & management - Set 1 (80 marks)

Answer all questions in the spaces provided.

0 1 . 1 Explain the difference between in-situ and ex-situ conservation and state when ex-situ conservation is needed.

[2 marks]

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0 1 . 2 Give **four** criteria used to select species for captive breeding and release programmes.

[4 marks]

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0 1 . 3 Explain why each of the following species would be difficult to keep or breed in captivity.

[4 marks]

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Blue whale .....

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Large blue butterfly .....

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0 2 The Hawaiian goose (nene) was reduced by hunting, habitat loss and introduced predators to about 30 birds. A captive breeding programme started with just seven individuals and has increased the wild population to over 2000.

0 2 . 1 Explain why starting the programme with seven individuals caused problems, and describe how these problems are managed.

[4 marks]

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0 2 . 2 Explain **three** reasons, other than gene pool size, why species kept in captivity may fail to breed.

[3 marks]

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0 2 . 3 Describe how artificial insemination and embryo transfer increase breeding success in captive mammals.

[3 marks]

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0 3 . 1 Explain the difference between hard and soft release and state the types of species for which each is suitable.

[3 marks]

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0 3 . 2 Describe **five** features of a suitable release site.

[5 marks]

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0 3 . 3 Explain the importance of post-release monitoring.

[2 marks]

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0 4 . 1 Describe how cloning could be used to conserve the northern white rhinoceros, and explain why success has been limited.

[3 marks]

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0 4 . 2 Explain how micro-propagation and cloning are used to conserve rare plants such as the Atlantic black poplar.

[3 marks]

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0 4 . 3 Describe the purpose and methods of the Millennium Seed Bank.

[4 marks]

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0 5 A conservation organisation is converting 700 ha of arable farmland on the coast into a wetland reserve for breeding and wintering birds (as at Wallasea Island).

0 5 . 1 Explain why habitat area and shape are important in the design of the reserve.

[4 marks]

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0 5 . 2 Explain how the reserve could be designed to provide a range of niches.

[3 marks]

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0 5 . 3 Explain why some species may need to be deliberately introduced while others colonise naturally.

[2 marks]

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0 5 . 4 Explain the value of a biological corridor linking the reserve to other wetlands.

[1 mark]

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0 6 . 1 Describe how each of the following abiotic factors can be managed to increase biodiversity in a nature reserve.

[6 marks]

Light .....

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Dissolved oxygen .....

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Nutrients .....

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0 6 . 2 Explain how biotic factors can be managed to increase the survival of a species. Give **four** examples.

[4 marks]

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0 7 Table 1 shows the number of plant species per m^2 in a chalk grassland reserve under different management, ten years after management began.

Table 1

Management	Mean plant species per m^2	Percentage of area covered by scrub / bushes
No management (abandoned)	8	55
Sheep grazing	24	5
Annual mowing (August)	19	0
Winter burning	14	10

0 7 . 1 Explain what is meant by a **plagioclimax** community and why chalk grassland is an example.

[2 marks]

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0 7 . 2 Use Table 1 to explain why grazing produced the highest plant diversity.

[4 marks]

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0 7 . 3 Explain why different grazing animals are chosen for different plagioclimax habitats.

[2 marks]

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0 7 . 4 Explain how burning maintains heathland.

[2 marks]

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0 8 . 1 Describe coppicing and explain how it increases the biodiversity of a woodland.

[4 marks]

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0 8 . 2 Explain how the reintroduction of grey wolves to Yellowstone National Park increased biodiversity.

[4 marks]

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0 8 . 3 Explain why culling or removing species may be necessary in habitat management.

[2 marks]

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END OF QUESTIONS

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